Interactive Structure Learning with Structural Query-by-Committee

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Abstract

In this work, we introduce *interactive structure learning*, a framework that unifies many different interactive learning tasks. We present a generalization of the *query-by-committee* active learning algorithm for this setting, and we study its consistency and rate of convergence, both theoretically and empirically, with and without noise.

1 Introduction

We introduce *interactive structure learning*, an abstract problem that encompasses many interactive learning tasks that have traditionally been studied in isolation, including active learning of binary classifiers, interactive clustering, interactive embedding, and active learning of structured output predictors. These problems include variants of both supervised and unsupervised tasks, and allow many different types of feedback, from binary labels to must-link/cannot-link constraints to similarity assessments to structured outputs. Despite these surface differences, they conform to a common template that allows them to be fruitfully unified.

In interactive structure learning, there is a space of items $\mathcal{X}$—for instance, an input space on which a classifier is to be learned, or points to cluster, or points to embed in a metric space—and the goal is to learn a *structure* on $\mathcal{X}$, chosen from a family $\mathcal{G}$. This set $\mathcal{G}$ could consist, for example, of all linear classifiers on $\mathcal{X}$, or all hierarchical clusterings of $\mathcal{X}$, or all knowledge graphs on $\mathcal{X}$. There is a target structure $g^* \in \mathcal{G}$ and the hope is to get close to this target. This is achieved by combining a loss function or prior on $\mathcal{G}$ with interactive feedback from an expert.

We allow this interaction to be fairly general. In most interactive learning work, the dominant paradigm has been *question-answering*: the learner asks a question (like "what is the label of this point x?") and the expert provides the answer. We allow a more flexible protocol in which the learner provides a constant-sized *snapshot* of its current structure and asks whether it is correct ("does the clustering, restricted to these ten points, look right?"). If the snapshot is correct, the expert accepts it; otherwise, the expert fixes some part of it. This type of feedback, first studied in generality by [15], can be called *partial correction*. It is a strict generalization of question-answering, and as we explain in more detail below, it allows more intuitive interactions in many scenarios.

In Section 3, we present *structural query-by-committee*, a simple algorithm that can be used for any instance of interactive structure learning. It is a generalization of the well-known query-by-committee (QBC) algorithm [33, 16], and operates, roughly, by maintaining a posterior distribution over structures and soliciting feedback on snapshots on which there is high uncertainty. We also introduce an adaptation of the algorithm that allows convex loss functions to handle the noise. This helps computational complexity in some practical settings, most notably when $\mathcal{G}$ consists of linear functions, and also makes it possible to efficiently kernelize structural QBC.

In Section 4, we show that structural QBC is guaranteed to converge to the target g^*, even when the expert's feedback is noisy. In the appendix, we give rates of convergence in terms of a *shrinkage*

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coefficient, present experiments on a variety of interactive learning tasks, and give an overview of related work.

2 Interactive structure learning

The space of possible interactive learning schemes is large and mostly unexplored. We can get a sense of its diversity from a few examples. In *active learning* [32], a machine is given a pool of unlabeled data and adaptively queries the labels of certain data points. By focusing on informative points, the machine may learn a good classifier using fewer labels than would be needed in a passive setting.

Sometimes, the labels are complex structured objects, such as parse trees for sentences or segmentations of images. In such cases, providing an entire label is time-consuming and it is easier if the machine simply suggests a label (such as a tree) and lets the expert either accept it or correct some particularly glaring fault in it. This is interaction with *partial correction*. It is more general than the *question-answering* usually assumed in active learning, and more convenient in many settings.

Interaction has also been used to augment *unsupervised* learning. Despite great improvements in algorithms for clustering, topic modeling, and so on, the outputs of these procedures are rarely perfectly aligned with the user's needs. The problem is one of underspecification: there are many legitimate ways to organize complex high-dimensional data, and no algorithm can magically guess which a user has in mind. However, a modest amount of interaction may help overcome this issue. For instance, the user can iteratively provide `must-link` and `cannot-link` constraints [37] to edit a flat clustering, or *triplet* constraints to edit a hierarchy [36].

These are just a few examples of interactive learning that have been investigated. The true scope of the settings in which interaction can be integrated is immense, ranging from structured output prediction to metric learning and beyond. In what follows, we aim to provide a unifying framework to address this profusion of learning problems.

2.1 The space of structures

Let $\mathcal{X}$ be a set of data points. This could be a pool of unlabeled data to be used for active learning, or a set of points to be clustered, or an entire instance space on which a metric will be learned.

We wish to learn a *structure* on $\mathcal{X}$, chosen from a class $\mathcal{G}$. This could, for instance, be the set of all labelings of $\mathcal{X}$ consistent with a function class $\mathcal{F}$ of classifiers (binary, multiclass, or with complex structured labels), or the set of all partitions of $\mathcal{X}$, or the set of all metrics on $\mathcal{X}$. Of these, there is some target $g^* \in \mathcal{G}$ that we wish to attain.

Although interaction will help choose a structure, it is unreasonable to expect that interaction alone could be an adequate basis for this choice. For instance, pinpointing a particular clustering over n points requires $\Omega(n)$ must-link/cannot-link constraints, which is an excessive amount of interaction when n is large.

To bridge this gap, we need a prior or a loss function over structures. For instance, if $\mathcal{G}$ consists of flat k-clusterings, then we may prefer clusterings with low k-means cost. If $\mathcal{G}$ consists of linear separators, then we may prefer functions with small norm $\|g\|$. In the absence of interaction, the machine would simply pick the structure that optimizes the prior or cost function. In this paper, we assume that this preference is encoded as a prior distribution π over $\mathcal{G}$.

We emphasize that although we have adopted a Bayesian formulation, there is no assumption that the target structure g^* is actually drawn from the prior.

2.2 Feedback

We consider schemes in which each individual round of interaction is not expected to take too long. This means, for instance, that the expert cannot be shown an entire clustering, of unrestricted size, and asked to comment upon it. Instead, he or she can only be given a small *snapshot* of the clustering, such as its restriction to 10 elements. The feedback on this snapshot will be either be to accept it, or to provide some constraint that fixes part of it.

In order for this approach to work, it is essential that structures be *locally checkable*: that is, g corresponds to the target g^* if and only if every snapshot of g is satisfactory.

When g is a clustering, for instance, the snapshots could be restrictions of g to subsets $S \subseteq \mathcal{X}$ of some fixed size s. Technically, it is enough to take $s = 2$, which corresponds to asking the user questions of the form 'Do you agree with having `zebra` and `giraffe` in the same cluster?" From the viewpoint of human-computer interaction, it might be preferable to use larger subsets (like $s = 5$ or $s = 10$), with questions such as "Do you agree with the clustering $\{\texttt{zebra}, \texttt{giraffe}, \texttt{dolphin}\}, \{\texttt{whale}, \texttt{seal}\}$?" Larger substructures provide more context and are more likely to contain glaring faults that the user can easily fix (`dolphin` and `whale` must go together). In general, we can only expect the user to provide partial feedback in these cases, rather than fully correcting the substructure.

2.3 Snapshots

Perhaps the simplest type of snapshot of a structure g is *the restriction of g to a small number of points*. We start by discussing this case, and later present a generalization.

2.3.1 Projections

For any $g \in \mathcal{G}$ and any subset $S \subseteq \mathcal{X}$ of size $s = O(1)$, let $g|_S$ be a suitable notion of the restriction of g to S, which we will sometimes call the *projection* of g onto S. For instance:

- $\mathcal{G}$ is a set of classifiers on $\mathcal{X}$. We can take $s = 1$ and let $g|_x$ be $(x, g(x))$ for any $x \in \mathcal{X}$.
- $\mathcal{G}$ is a set of partitions (flat clusterings) of $\mathcal{X}$. For a set $S \subseteq \mathcal{X}$ of size $s \geq 2$, let $g|_S$ be the induced partition on just the points S.

As discussed earlier, it will often be helpful to pick projections of size larger than the minimal possible s. For clusterings, for instance, any $s \geq 2$ satisfies local checkability, but human feedback might be more effective when $s = 10$ than when $s = 2$. Thus, in general, the queries made to the expert will consist of snapshots (projections of size $s = 10$, say) that can in turn be decomposed further into *atomic units* (projections of size 2).

2.3.2 Atomic decompositions of structures

Now we generalize the notion of projection to other types of snapshots and their atomic units.

We will take a functional view of the space of structures $\mathcal{G}$, in which each structure g is specified by its "answers" to a set of *atomic questions* $\mathcal{A}$. For instance, if $\mathcal{G}$ is the set of partitions of $\mathcal{X}$, then $\mathcal{A} = \binom{\mathcal{X}}{2}$, with $g(\{x, x'\}) = 1$ if g places x, x' in the same cluster and 0 otherwise.

The queries made during interaction can, in general, be composed of multiple atomic units, and feedback will be received on at least one of these atoms. Formally, let $\mathcal{Q}$ be the space of queries. In the partitioning example, this might be $\binom{\mathcal{X}}{10}$. The relationship between $\mathcal{Q}$ and $\mathcal{A}$ is captured by the following requirements:

- Each $q \in \mathcal{Q}$ can be decomposed as a set of atomic questions $A(q) \subseteq \mathcal{A}$, and we write $g(q) = \{(a, g(a)) : a \in A(q)\}$. In the partitioning example, $A(q)$ is the set of all pairs in q.
- The user accepts $g(q)$ if and only if g satisfactorily answers every atomic question in q, that is, if and only if $g(a) = g^*(a)$ for all $a \in A(q)$.

2.4 Summary of framework

To summarize, interactive structure learning has two key components:

- *A reduction to multiclass classifier learning.* We view each structure $g \in \mathcal{G}$ as a function on atomic questions $\mathcal{A}$. Thus, learning a good structure is equivalent to picking one whose labels $g(a)$ are correct.
- *Feedback by partial correction.* For practical reasons we consider broad queries, from a set $\mathcal{Q}$, where each query can be decomposed into atomic questions, allowing for partial corrections. This decomposition is given by the function $A : \mathcal{Q} \to 2^{\mathcal{A}}$.

Algorithm 1 STRUCTURAL QBC

Input: Distribution[1] ν over query space $\mathcal{Q}$ and initial prior distribution π_o over $\mathcal{G}$
Output: Posterior distribution π_t over $\mathcal{G}$
for $t = 1, 2, \ldots$ **do**
  Draw $g_t \sim \pi_{t-1}$
  while Next query q_t has not been chosen **do**
    Draw $q \sim \nu$ and $g, g' \sim \pi_{t-1}$
    With probability $d(g, g'; q)$: take $q_t = q$
  end while
  Show user q_t and $g_t(q_t)$ and receive feedback in form of pairs (a_t, y_t)
  Update posterior: $\pi_t(g) \propto \pi_{t-1}(g) \exp(-\beta \cdot \mathbf{1}(g(a_t) \neq y_t))$
end for

The reduction to multiclass classification immediately suggests algorithms that can be used in the interactive setting. We are particular interested in *adaptive* querying, with the aim of finding a good structure with minimal interaction. Of the many schemes available for binary classifiers, one that appears to work well in practice and has good statistical properties is *query-by-committee* [33, 16]. It is thus a natural candidate to generalize to the broader problem of structure learning.

3 Structural QBC

Query-by-committee, as originally analyzed by [16], is an active learning algorithm for binary classification in the noiseless setting. It uses a prior probability distribution π over its classifiers and keeps track of the current version space, i.e. the classifiers consistent with the labeled data seen so far. At any given time, the next query is chosen as follows:

- Repeat:
  - Pick $x \in \mathcal{X}$ at random
  - Pick classifiers h, h' at random from π restricted to the current version space
  - If $h(x) \neq h'(x)$: halt with x as the query

In our setting, the feedback at time t is the answer y_t to some atomic question $a_t \in \mathcal{A}$, and we can define the resulting version space to be $\{g \in \mathcal{G} : g(a_{t'}) = y_{t'} \text{ for all } t' \leq t\}$. The immediate generalization of QBC would involve picking a query $q \in \mathcal{Q}$ at random (or more generally, drawn from some query distribution ν), and then choosing it if g, g' sampled from π restricted to our version space happen to disagree on it. But this is unlikely to work well, because the answers to queries are no longer binary labels but mini-structures. As a result, g, g' are likely to disagree on minor details even when the version space is quite small, leading to excessive querying. To address this, we will use a more refined notion of the difference between $g(q)$ and $g'(q)$:

$$d(g, g'; q) \;=\; \frac{1}{|A(q)|} \sum_{a \in A(q)} \mathbf{1}[g(a) \neq g'(a)].$$

In words, this is the fraction of atomic subquestions of q on which g and g' disagree. It is a value between 0 and 1, where higher values mean that $g(q)$ differs significantly from $g'(q)$. Then we will query q with probability $d(g, g'; q)$.

3.1 Accommodating noisy feedback

We are interested in the noisy setting, where the user's feedback may occasionally be inconsistent with the target structure. In this case, the notion of a version space is less clear-cut. Our modification is very simple: the feedback at time t, say (a_t, y_t), causes the posterior to be updated as follows:

$$\pi_t(g) \;\propto\; \pi_{t-1}(g) \exp(-\beta \cdot \mathbf{1}[g(a_t) \neq y_t]). \tag{1}$$

[1] In the setting where $\mathcal{Q}$ is finite, a reasonable choice of ν would be uniform over $\mathcal{Q}$.

Here $\beta > 0$ is a constant that controls how aggressively errors are punished. In the noiseless setting, we can take $\beta = \infty$ and recover the original QBC update. Even with noise, however, this posterior update still enjoys nice theoretical properties. The full algorithm is shown in Algorithm 1.

3.2 Uncertainty and informative queries

What kinds of queries will structural QBC make? To answer this, we first quantify the *uncertainty* in the current posterior about a particular query or atom. Define the uncertainty of atom $a \in \mathcal{A}$ under distribution distribution π as $u(a; \pi) = \Pr_{g,g' \sim \pi}(g(a) \neq g'(a))$ and $u(q; \pi)$ as the average uncertainty of its atoms $A(q)$. These values lie in the range $[0, 1]$.

The probability that a particular query $q \in \mathcal{Q}$ is chosen in round t by structural QBC is proportional to $\nu(q)u(q; \pi_{t-1})$. Thus, queries with higher uncertainty under the current posterior are more likely to be chosen. As the following lemma demonstrates, getting feedback on uncertain atoms eliminates, or down-weights in the case of noisy feedback, many structures inconsistent with g^*.

Lemma 1. *For any distribution π over $\mathcal{G}$, we have $\pi(\{g : g(a) \neq y\}) \geq u(a; \pi)/2$.*

The proof of Lemma 1 is deferred to the appendix. This gives some intuition for the query selection criterion of structural QBC, and will later be used in the proof of consistency.

3.3 General loss functions

The update rule for structural QBC, equation (1), results in a posterior of the form $\pi_t(g) \propto \pi(g) \exp(-\beta \cdot \#(\text{mistakes made by } g))$, which may be difficult to sample from. Thus, we consider a broader class of updates,

$$\pi_t(g) \;\propto\; \pi_{t-1}(g) \exp(-\beta \cdot \ell(g(a_t), y_t)), \tag{2}$$

where $\ell(\cdot, \cdot)$ is a general loss function. In the special case where $\mathcal{G}$ consists of linear functions and ℓ is convex, π_t will be a log-concave distribution, which allows for efficient sampling [28]. We will show that this update also enjoys nice theoretical properties, albeit under different noise conditions.

To formally specify this setting, let $\mathcal{Y}$ be the space of answers to atomic questions $\mathcal{A}$, and suppose that structures in $\mathcal{G}$ generate values in some prediction space $\mathcal{Z} \subseteq \mathbb{R}^d$. That is, each $g \in \mathcal{G}$ is a function $g : \mathcal{A} \to \mathcal{Z}$, and any output $z \in \mathcal{Z}$ gets translated to some prediction in $\mathcal{Y}$. The loss associated with predicting z when the true answer is y is denoted $\ell(z, y)$. Here are some examples:

- $0 - 1$ loss. $\mathcal{Z} = \mathcal{Y}$ and $\ell(z, y) = \mathbf{1}(y \neq z)$.
- Squared loss. $\mathcal{Y} = \{-1, 1\}$, $\mathcal{Z} = [-B, B]$, and $\ell(z, y) = (y - z)^2$.
- Logistic loss. $\mathcal{Y} = \{-1, 1\}$, $\mathcal{Z} = [-B, B]$ for some $B > 0$, and $\ell(z, y) = \ln(1 + e^{-yz})$.

When moving from a discrete to a continuous prediction space, it becomes very possible that the predictions, on a particular atom, of two randomly chosen structures will be close but not perfectly aligned. Thus, instead of checking strict equality of these predictions, we need to modify our querying strategy to take into account the distance between them. To this end, we will use the normalized average squared Euclidean distance:

$$d^2(g, g'; q) \;=\; \frac{1}{|A(q)|} \sum_{a \in A(q)} \frac{\|g(a) - g'(a)\|^2}{D}$$

where $D = \max_{a \in \mathcal{A}} \max_{g,g' \in \mathcal{G}} \|g(a) - g'(a)\|^2$. Note that $d^2(g, g'; q)$ is a value between 0 and 1. We treat it as a probability, in exactly the same way we used $d(g, g'; q)$ in the 0-1 loss setting.

In the 0-1 loss setting, structural QBC chooses queries proportional to their uncertainty. What queries will structural QBC make in the general loss setting? Define the variance of $a \in \mathcal{A}$ under π as

$$\text{var}(a; \pi) = \frac{1}{2} \sum_{g,g' \in \mathcal{G}} \pi(g)\, \pi(g')\, \|g(a) - g'(a)\|^2$$

and $\text{var}(q; \pi)$ as the average variance of its atoms $A(q)$. Then the probability that structural QBC chooses $q \in \mathcal{Q}$ at step t is proportional to $\nu(q)\text{var}(q; \pi_{t-1})$ in the general loss setting.

Algorithm 2 ROBUST QUERY SELECTION

Input: Fixed set of queries $q_1, \ldots, q_m \in \mathcal{Q}$, current distribution π over $\mathcal{G}$
Output: Query q_i
Initial shrinkage estimate: $\widehat{u}_o = 1/2$
for $t = 0, 1, 2, \ldots$ **do**
  Draw $g_1, g_1', \ldots, g_{n_t}, g_{n_t}' \sim \pi$
  If there exists q_j such that $\frac{1}{n_t}\sum_{i=1}^{n_t} d(g_i, g_i'; q_j) \geq \widehat{u}_t$ then we halt and query q_j
  Otherwise, let $\widehat{u}_{t+1} = \widehat{u}_t/2$.
end for

3.4 Kernelizing structural QBC

Consider the special case where $\mathcal{G}$ consists of linear functions, i.e. $\mathcal{G} = \{g_w(x) = \langle x, w\rangle \,:\, w \in \mathbb{R}^d\}$. As mentioned above, when our loss function is convex, the posteriors we encounter are log-concave, and thus efficiently samplable. But what if we want a more expressive class than linear functions? To address this, we will resort to kernels.

Gilad-Bachrach et al. [17] investigated the use of kernels in QBC. In particular, they observed that QBC does not actually need samples from the prior restricted to the current version space. Rather, given a candidate query x, it is enough to be able to sample from the distribution the posterior induces over the labelings of x. Although their work was in the realizable binary setting, this observation still applies to our setting.

Given a feature mapping $\phi : \mathcal{X} \to \mathbb{R}^d$, our posterior update becomes $\pi_t(g_w) \propto \pi_{t-1}(g_w)\exp\left(-\beta\ell(\langle\phi(x_t), w\rangle, y_t)\right)$. As the following lemma shows, when $\ell(\cdot,\cdot)$ is the squared-loss and our prior is Gaussian, the predictions of the posterior have a univariate normal distribution.

Lemma 2. *Suppose $\pi = \mathcal{N}(0, \sigma_o^2 I_d)$, $\ell(\cdot,\cdot)$ is the squared-loss, and we have observed $(x_1, y_1), \cdots, (x_t, y_t)$. If $g_w \sim \pi_t$, then $\langle w, \phi(x)\rangle \sim \mathcal{N}(\mu, \sigma^2)$ where*

$$\mu = 2\sigma_o^2\beta\kappa^T\left(I_t - \Sigma_o K\right)y \qquad \textit{and} \qquad \sigma^2 = \sigma_o^2\left(\phi(x)^T\phi(x) - \kappa^T\Sigma_o\kappa\right)$$

for $K_{ij} = \langle\phi(x_i), \phi(x_j)\rangle$, $\kappa_i = \langle\phi(x_i), \phi(x)\rangle$, and $\Sigma_o = \left(\frac{1}{2\sigma^2\beta}I_t + K\right)^{-1}$.

The proof is deferred to the appendix. The important observation here is that all the quantities involving the feature mapping in Lemma 2 are inner products. Thus we never need to explicitly construct any feature vectors.

3.5 Reducing the randomness in structural QBC

It is easy to see that the query selection procedure of structural QBC is a rejection sampler where each query q is chosen with probability proportional to $\nu(q)u(q;\pi_t)$ (in the case of the 0-1 loss) or $\nu(q)\mathrm{var}(q;\pi_t)$ (for general losses). However, it is possible for the rejection rate to be quite high, even when there are many queries that have much higher uncertainty or variance than the rest. To circumvent this issue, we introduce a 'robust' version of structural QBC, wherein many candidate queries are sampled, and the query that has the highest uncertainty or variance is chosen.

In the 0-1 loss case, we can estimate the uncertainty of a candidate query q by drawing many pairs $g_1, g_1', \ldots, g_n, g_n' \sim \pi_t$ and using the unbiased estimator $\widehat{u}(q;\pi_t) := \frac{1}{n}\sum_{i=1}^{n} d(g_i, g_i'; q)$.

Unfortunately, the number of structures we need to sample in order to identify the most uncertain query depends on its uncertainty, which we do not know a priori. To circumvent this difficulty, we can use the halving procedure shown in Algorithm 2. If the appropriate number of structures are sampled at each round t, on the order of $O((1/u_t^2)\log(m\log(1/u_o)))$ for some crude lower bound u_o on the highest uncertainty, then with high probability this procedure terminates with a candidate query whose uncertainty is within a constant factor of the highest uncertainty [35].

4 Consistency of structural QBC

In this section, we look at a typical setting in which there is a finite but possibly very large pool of candidate questions $\mathcal{Q}$, and thus the space of structures $\mathcal{G}$ is effectively finite. Let $g^* \in \mathcal{G}$ be the target structure, as before. Our goal in this setting is to demonstrate the *consistency* of structural QBC, meaning that $\lim_{t\to\infty} \pi_t(g^*) = 1$ almost surely. To do so, we formalize our setting. Note that the random outcomes during time step t of structural QBC consist of the query q_t, the atomic question a_t that the expert chooses to answer (pick one at random if the expert answers several of them), and the response y_t to a_t. Let $\mathcal{F}_t$ denote the sigma-field of all outcomes up to, and including, time t.

4.1 Consistency under 0-1 loss

In order to prove consistency, we will have to make some assumptions about the feedback we receive from a user. For query $q \in \mathcal{Q}$ and atomic question $a \in A(q)$, let $\eta(y|a, q)$ denote the conditional probability that the user answers y to atomic question a, in the context of query q. Our first assumption is that the single most likely answer is $g^*(a)$.

Assumption 1. *There exists $0 < \lambda \leq 1$ such that $\eta(g^*(a)|a, q) - \eta(y|a, q) \geq \lambda$ for all $q \in \mathcal{Q}$ and $a \in A(q)$ and all $y \neq g^*(a)$.*

(We will use the convention $\lambda = 1$ for the noiseless setting.) In the learning literature, Assumption 1 is known as Massart's bounded noise condition [2].

The following lemma, whose proof is deferred to the appendix, demonstrates that under Assumption 1, the posterior probability of g^* increases in expectation with each query, as long as the β parameter of the update rule in equation (1) is small enough relative to λ.

Lemma 3. *Fix any t and suppose the expert provides an answer to atomic question $a_t \in A(q_t)$ at time t. Let $\gamma_t = \pi_{t-1}(\{g \in \mathcal{G} : g(a_t) = g^*(a_t)\})$. Define Δ_t by:*

$$\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}, q_t, a_t\right] \;=\; (1-\Delta_t)\frac{1}{\pi_{t-1}(g^*)},$$

Under Assumption 1, Δ_t can be lower-bounded as follows:

(a) If $\lambda = 1$ (noiseless setting), $\Delta_t \geq (1-\gamma_t)(1-e^{-\beta})$.

(b) For $\lambda \in (0, 1)$, if $\beta \leq \lambda/2$, then $\Delta_t \geq \beta\lambda(1-\gamma_t)/2$.

To understand the requirement $\beta = O(\lambda)$, consider an atomic question on which there are just two possible labels, 1 and 2, and the expert chooses these with probabilities p_1 and p_2, respectively. If the correct answer according to g^* is 1, then $p_1 \geq p_2 + \lambda$ under Assumption 1. Let $\mathcal{G}_2$ denote structures that answer 2.

- With probability p_1, the expert answers 1, and the posterior mass of $\mathcal{G}_2$ is effectively multiplied by $e^{-\beta}$.
- With probability p_2, the expert answers 2, and the posterior mass of $\mathcal{G}_2$ is effectively multiplied by e^{β}.

The second outcome is clearly undesirable. In order for it to be counteracted, in expectation, by the first, β must be small relative to p_1/p_2. The condition $\beta \leq \lambda/2$ ensures this.

Lemma 3 does not, in itself, imply consistency. It is quite possible for $1/\pi_t(g^*)$ to keep shrinking but not converge to 1. Imagine, for instance, that the input space has two parts to it, and we keep improving on one of them but not the other. What we need is, first, to ensure that the queries q_t capture some portion of the uncertainty in the current posterior, and second, that the user chooses atoms that are at least slightly informative. The first condition is assured by the SQBC querying strategy. For the second, we need an assumption.

Assumption 2. *There is some minimum probability $p_o > 0$ for which the following holds. If the user is presented with a query q and a structure $g \in \mathcal{G}$ such that $g(q) \neq g^*(q)$, then with probability at least p_o the user will provide feedback on some $a \in A(q)$ such that $g(a) \neq g^*(a)$.*

Assumption 2 is one way of avoiding scenarios in which a user never provides feedback on a particular atom a. In such a pathological case, we might not be able to recover $g^*(a)$, and thus our posterior will always put some probability mass on structures that disagree with g^* on a.

The following lemma gives lower bounds on $1 - \gamma_t$ under Assumption 2.

Lemma 4. *Suppose that $\mathcal{G}$ is finite and the user's feedback obeys Assumption 2. Then there exists a constant $c > 0$ such that for every round t*

$$\mathbb{E}[1 - \gamma_t \,|\, \mathcal{F}_{t-1}] \;\geq\; c\,\pi_{t-1}(g^*)^2(1 - \pi_{t-1}(g^*))^2$$

where $\gamma_t = \pi_{t-1}(\{g \in \mathcal{G} : g(a_t) = g^*(a_t)\})$ and a_t is the atom the user provides feedback on.

Together, Lemmas 3 and 4 show that the sequence $\frac{1}{\pi_t(g^*)}$ is a positive supermartingale that decreases in expectation at each round by an amount that depends on $\pi_t(g^*)$. The following lemma tells us exactly when such stochastic processes can be guaranteed to converge.

Lemma 5. *Let $f : [0, 1] \to \mathbb{R}_{\geq 0}$ be a continuous function such that $f(1) = 0$ and $f(x) > 0$ for all $x \in (0, 1)$. If*

$$\mathbb{E}\left[\frac{1}{\pi_t(g^*)} \,\middle|\, \mathcal{F}_{t-1}\right] \leq \frac{1}{\pi_{t-1}(g^*)} - f(\pi_{t-1}(g^*))$$

for each $t \in \mathbb{N}$, then $\pi_t(g^*) \to 1$ almost surely.

As a corollary, we see that structural QBC is consistent.

Theorem 6. *Suppose that $\mathcal{G}$ is finite, and Assumptions 1 and 2 hold. If $\pi(g^*) > 0$, then $\pi_t(g^*) \to 1$ almost surely under structural QBC's query strategy.*

We provide a proof of Theorem 6 in the appendix, where we also provide rates of convergence.

4.2 Consistency under general losses

We now turn to analyzing structural QBC with general losses. As before, we will need to make some assumptions. The first is that the loss function is well-behaved.

Assumption 3. *The loss function is bounded, $0 \leq \ell(z, y) \leq B$, and Lipschitz in its first argument, i.e. $\ell(z, y) - \ell(z', y) \leq C\|z - z'\|$, for some constants $B, C > 0$.*

It is easily checked that this assumption holds for the three loss functions we mentioned earlier.

In the case of 0-1 loss, we assumed that for any atomic question a, the correct answer $g^*(a)$ would be given with higher probability than any incorrect answer. We now formulate an analogous assumption for the case of more general loss functions. Recall that $\eta(\cdot|a)$ is the conditional probability distribution over the user's answers to $a \in \mathcal{A}$ (we can also allow η to also depend upon the context q, as we did before; here we drop the dependence for notational convenience). The expected loss incurred by $z \in \mathcal{Z}$ on this atom is thus

$$L(z, a) = \sum_y \eta(y|a)\,\ell(z, y).$$

We will require that for any atomic question a, this expected loss is minimized when $z = g^*(a)$, and predicting any other z results in expected loss that grows with the distance between z and $g^*(a)$.

Assumption 4. *There exists a constant $\lambda > 0$ such that $L(z, a) - L(g^*(a), a) \;\geq\; \lambda\|z - g^*(a)\|^2$ for any atomic question $a \in \mathcal{A}$ and any $z \in \mathcal{Z}$.*

Let's look at some concrete settings:

- $0 - 1$ loss with $\mathcal{Y} = \mathcal{Z} = \{0, 1\}$. Assumption 4 is equivalent to Assumption 1.
- Squared loss with $\mathcal{Y} = \{-1, 1\}$ and $\mathcal{Z} \subset \mathbb{R}$. Assumption 4 is satisfied when $g^*(a) = \mathbb{E}[y|a]$ and $\lambda = 1$.
- Logistic loss with $\mathcal{Y} = \{-1, 1\}$ and $\mathcal{Z} = [-B, B]$. For $a \in A$, let $p = \eta(1|a)$. Assumption 4 is satisfied when $g^*(a) = \ln \frac{p}{1-p}$ and $\lambda = \frac{2e^{2B}}{(1+e^B)^4}$.

From these examples, it is clear that requiring $g^*(a)$ to be the minimizer of $L(z, a)$ is plausible if $\mathcal{Z}$ is a discrete space but much less so if $\mathcal{Z}$ is continuous. In general, we can only hope that this holds approximately. With this caveat in mind, we stick with Assumption 4 as a useful but idealized mathematical abstraction.

With these assumptions in place, the following theorem guarantees the consistency of structural QBC under general losses. Its proof is deferred to the appendix.

Theorem 7. *Suppose we are in the general loss setting, $\mathcal{G}$ is finite, and the user's feedback satisfies Assumptions 2, 3, and 4. If $\pi(g^*) > 0$, then $\pi_t(g^*) \to 1$ almost surely.*

5 Conclusion

In this work, we introduced interactive structure learning, a generic framework for learning structures under partial correction feedback. This framework can be applied to any structure learning problem in which structures are in one-to-one correspondence with their answers to atomic questions. Thus, interactive structure learning may be viewed as a generalization of active learning, interactive clustering with pairwise constraints, interactive hierarchical clustering with triplet constraints, and interactive ordinal embeddings with quadruplet constraints.

On the algorithmic side, we introduced structural QBC, a generalization of the classical QBC algorithm to the interactive structure learning setting. We demonstrated that this algorithm is consistent, even in the presence of noise, provided that we can sample from a certain natural posterior. In the appendix, we also provided rates of convergence. Because this posterior is often intractable to sample from, we also considered an alternative posterior based on convex loss functions that sometimes allows for efficient sampling. We showed that structural QBC remains consistent in this setting, albeit under different noise conditions.

In the appendix, we provide experiments on both interactive clustering and active learning tasks. On the interactive clustering side, these experiments demonstrate that even when the prior distribution places relatively low mass on the target clustering, structural QBC is capable of recovering a low-error clustering with relatively few rounds of interaction. In contrast, these experiments also show that random corrections are not quite as useful. On the active learning side, there are experiments demonstrating the good empirical performance of structural QBC using linear classifiers with the squared-loss posterior update, with and without kernelization.

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A Related work

In this section, we discuss some areas of work related to the current work.

Active learning. In pool-based active learning, a learner starts with a large collection, or pool, of unlabeled data and adaptively queries points for their labels. Several variations on this basic setting have been studied, but we illustrate just two of them here.

The first is the nonparameteric active learning setting in which the algorithm requests a few of the labels and then fills in the rest. Several algorithms exist for this setting, but they all share the same objective: query few data points and accurately label the rest. The strategies for this setting range from graph-based [39, 29, 10] to hierarchical clustering-based [13]. In general, the success of methods in this setting usually depends on some smoothness property of the underlying data distribution [8, 25].

The second setting of interest involves learning some fixed unknown concept belonging to a known concept class $\mathcal{H}$. Active learning strategies for this setting range from mellow [9, 14], where a label is requested for any point whose label cannot be inferred from those already queried, to aggressive, where only highly informative points are queried. Among the latter, the particular definition of 'informative' varies. In some strategies, like query-by-committee [33, 16] and generalized binary search [11, 19, 30], informative points are those that eliminate many hypotheses. Other strategies, like the splitting-index algorithm [12] and the related DBAL algorithm [35], query points that eliminate very different hypotheses.

The work presented here bears resemblance to both of these settings. Indeed, our definition of structure is general enough to accommodate arbitrary labelings of a data set, as in the nonparametric settings. However, the algorithms considered here are much more closely related to those considered in the parametric setting. Where this work most clearly departs from the classical active learning literature is the change from question-answering to partial correction feedback.

Additionally, recent works in this area have looked at bridging the gap between theory and practice by providing algorithms with theoretical guarantees that make use of well-studied algorithmic primitives. The two typical primitives considered in the literature are empirical risk minimization [7, 23] and sampling [18], both of which have been shown to exhibit efficient algorithms in certain cases of interest. The present work makes progress on demonstrating that efficient sampling algorithms from distributions of the form $e^{-\beta L(g)}$, where $L(g)$ is the empirical loss of a function g, leads to efficient active learning algorithms to find functions with small loss.

Interactive clustering. Another area relevant to this work is interactive clustering, which has also developed along several fronts.

One interactive clustering model considered in the literature allows users to split and merge clusters in the algorithm's current clustering until the target clustering has been found. In order to limit the load place on a user, split requests only specify that a cluster should be split, not how to split it. It is up to the algorithm to decide how to go about splitting the chosen cluster. Under certain assumptions, various algorithms have been shown to recover the target clustering while needing relatively few rounds of interaction [6, 4, 3].

Another approach to interactive clustering, and the one that falls within the scope of this work, is the clustering with constraints framework [37, 36, 1]. As discussed in Section 2, constraints in both flat and hierarchical clustering can be written using in our interactive structure learning framework. It is less clear, however, how to incorporate split and merge feedback into our model. It thus remains an interesting open problem to marry constraint-based interaction with higher-level feedback such as cluster splits and merges.

Learning with alternative feedback. Recently, researchers have investigated the use of additional forms of feedback, beyond simply providing labels, to assist in supervised learning tasks. Two forms of feedback that have been considered are pairwise comparisons [24, 38], which indicate which of two points is more likely to be labeled positive, and feature feedback [31], in which a labeler indicates a particularly relevant feature. Both of these forms of feedback have been shown to be useful for supervised and active learning under certain conditions. It is an interesting open question as to whether such side-information feedback can be incorporated in the more generic interactive structure learning framework considered here.

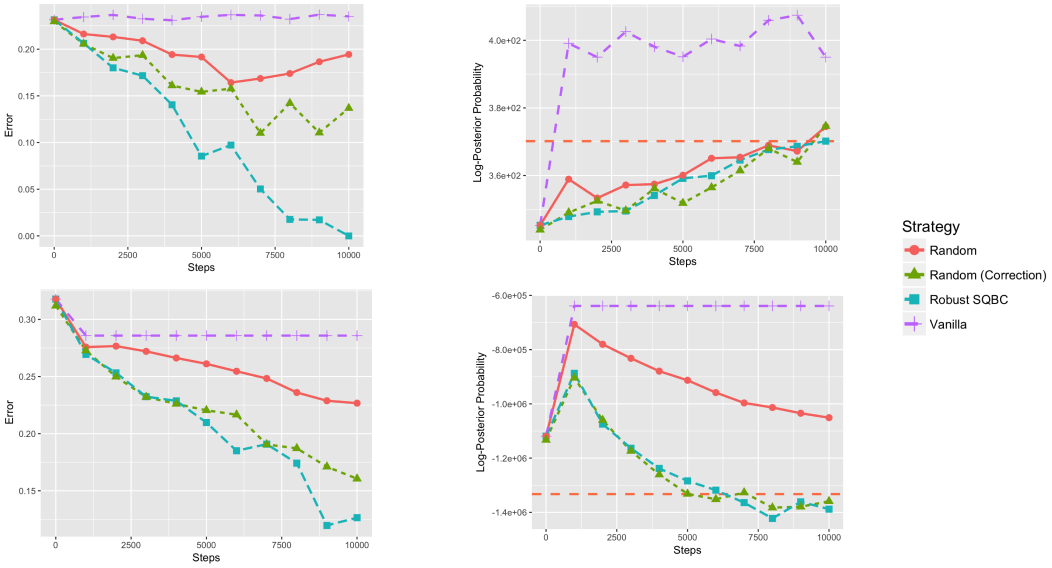

Figure 1: Mixture of Gaussians clustering experiments. x-axis corresponds to full passes of Gibbs sampler. Dashed orange line corresponds to log-posterior of target clustering. *Top*: Iris dataset. *Bottom*: Wine dataset.

B Experiments

We now turn to our experiments with structural QBC in a variety of applications.

B.1 Clustering with mixtures of Gaussians

Consider the following Bayesian generative model for a mixture of k spherical Gaussians.

$$(w_1, \ldots, w_k) \sim \text{Dirichlet}(\alpha) \qquad z_i \mid w \sim \text{Categorical}(w_1, \ldots, w_k)$$
$$\mu^{(1)}, \ldots, \mu^{(k)} \sim \mathcal{N}(\mu_o, \sigma_o^2 I_d) \qquad x^{(i)} \mid z, \mu \sim \mathcal{N}(\mu_{z_i}, \sigma^2 I_d)$$

Here, $x^{(1)}, \ldots, x^{(n)}$ are the observed data points; α, μ_o, σ_o^2, and σ^2 are known hyper-parameters; and the rest are unobserved latent variables. Our goal is to find a target clustering with the Bayesian posterior distribution $\Pr(z \mid x)$ acting as our prior π over clusterings.

In each of our experiments, we ran the robust version of structured QBC (denoted 'Robust SQBC') where the candidate queries are subsets of size ten and feedback consists of a random correction to the proposed subset clustering. To sample from this posterior, we used the collapsed Gibbs sampler. Although this Markov chain can take exponential time to mix [34], we found that running structural QBC with these samples often led to fast convergence to the underlying clustering.

We compared against three baseline strategies: a random strategy (denoted 'Random') wherein feedback was provided on randomly sampled pairs of point, a random correction strategy (denoted 'Random (correction)') wherein feedback was a random correction to a proposed clustering of ten random points, and a no-feedback strategy (denoted 'Vanilla') in which we simply ran an unconstrained Gibbs sampler. For all strategies, we measured the clustering distance from the current clustering to the target clustering and the log-posterior probability after every pass of the Gibbs sampler. For the feedback strategies, a constraint was added every 50 passes of the Gibbs sampler. We ran experiments on the wine and iris datasets from the UCI machine learning repository [27]. The results are displayed in Figure 1.

B.2 Clustering with mixtures of Bernoulli products

Consider the following Bayesian generative model for a mixture of k Bernoulli product distributions:

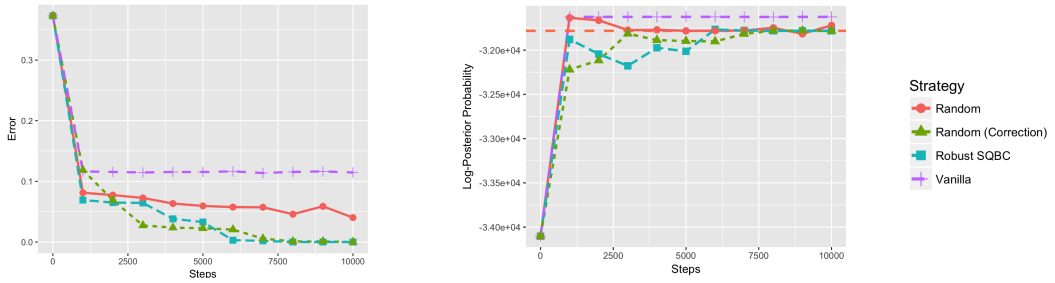

Figure 2: Mixture of Bernoullis clustering experiments. x-axis corresponds to full passes of Gibbs sampler. Dashed orange line corresponds to log-posterior of target clustering.

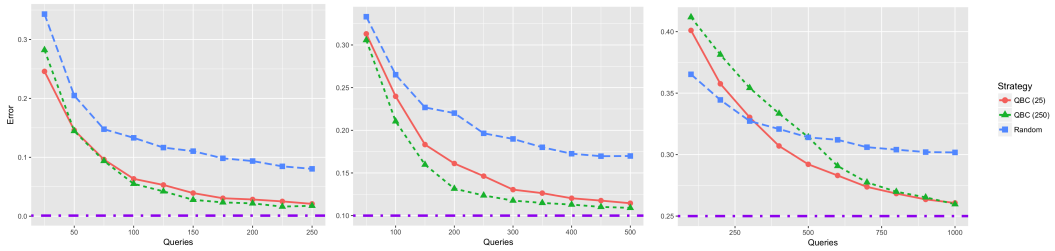

Figure 3: Linear separator simulations. From left to right classification noise: $p = 0.001$, $p = 0.1$, and $p = 0.25$. The dashed purple line is the level of classification noise. In the legend, QBC run with update β is shown as 'QBC (β).'

- Weight vector $(w_1, \ldots, w_k)$ is drawn from a symmetric Dirichlet distribution with parameter $\alpha > 0$
- Bias variables $a_j^{(i)} \in [0, 1]$ are drawn i.i.d. from Beta(β, γ) for $i = 1, \ldots, k$ and $j = 1, \ldots, d$
- For each data point $i = 1, \ldots, n$:
  - $z_i \in \{1, \ldots, k\}$ is drawn from Categorical$(w_1, \ldots, w_k)$
  - $x_j^{(i)} \in \{0, 1\}$ is drawn from Bern$(a_j^{(z_i)})$ for $j = 1, \ldots, d$

Here, $x^{(1)}, \ldots, x^{(n)}$ are the observed d-dimensional binary-valued vectors, α, β, and γ are known hyperparameters, and everything else is a latent variable,

As in the the mixtures of Gaussians experiments, we sought to find a target clustering with the Bayesian posterior distribution $\Pr(z \mid x)$ acting as our prior π over clusterings. We used the binarized version of the MNIST handwritten digit dataset [26]. Here we randomly sampled 50 images from each of the 0, 1, and 2 classes. The results are presented in Figure 2.

B.3 Linear separators

We also considered the setting of learning linear separators. In these experiments, we used QBC with a spherical prior distribution $\mathcal{N}(0, I_d)$ and the squared-loss posterior update.

When learning a linear separator under classification noise, there is a true linear separator $h^* \in \mathbb{R}^d$. When a point $x \in \mathbb{R}^d$ is queried, we observe sign$(\langle h^*, x \rangle)$ with probability $1 - p$ and $-$sign$(\langle h^*, x \rangle)$ with probability p.

In our simulations, we used various settings of both the noise level p and the aggressiveness of the posterior update β. In the low noise setting, we found that setting β large appears to be appropriate. But as the noise level grows, using a smaller β appears to be advantageous. In all settings, however, QBC outperforms random sampling. Figure 3 shows the results of these simulations.

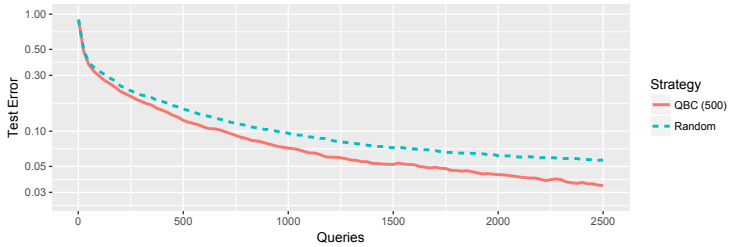

Figure 4: Kernelized MNIST experiment. QBC run with posterior update β is shown as 'QBC (β)' in the legend. Note that the test error axis is log-scale.

B.4 Kernel QBC experiments

We used the full MNIST handwritten digit dataset for our linear classification experiment. For this multiclass classification task, we used interactively trained a collection of ten one-vs-all classifiers using the kernelized least-squares approach outlined in Section 3. We used an RBF kernel $K(x, x') = \exp(-\gamma\|x - x'\|^2)$ with the choice of $\gamma = 0.001$. Because we are observing the true labels, we found that large values of β worked well for this task. The results are presented in Figure 4.

C Deferred Proofs

C.1 Proof of Lemma 1

Lemma 1. *For any distribution π over $\mathcal{G}$, we have $\pi(\{g : g(a) \neq y\}) \geq u(a; \pi)/2$.*

Proof. Suppose the possible answers to a are $y_1, y_2, \ldots$, and that these have probabilities $p_1 \geq p_2 \geq \cdots$ respectively under π. That is $p_i = \pi(\{g : g(a) = y_i\})$. Then

$$u(a; \pi) \;=\; 1 - \sum_i p_i^2.$$

Note that $\pi(\{g : g(a) \neq y\})$ is smallest when $y = y_1$. Thus, we have

$$1 - \pi(\{g : g(a) \neq y\}) \;\leq\; p_1 \;\leq\; \sqrt{1 - u(a; \pi)} \;\leq\; 1 - \frac{1}{2}u(a; \pi),$$

Rearranging gives us the lemma. □

C.2 Proof of Lemma 2

Our goal is to prove the following lemma.

Lemma 2. *Suppose $\pi = \mathcal{N}(0, \sigma_o^2 I_d)$, $\ell(\cdot, \cdot)$ is the squared-loss, and we have observed $(x_1, y_1), \cdots, (x_t, y_t)$. If $g_w \sim \pi_t$, then $\langle w, \phi(x)\rangle \sim \mathcal{N}(\mu, \sigma^2)$ where*

$$\mu = 2\sigma_o^2 \beta \kappa^T \left(I_t - \Sigma_o K\right) y \qquad \text{and} \qquad \sigma^2 = \sigma_o^2 \left(\phi(x)^T \phi(x) - \kappa^T \Sigma_o \kappa\right)$$

for $K_{ij} = \langle \phi(x_i), \phi(x_j)\rangle$, $\kappa_i = \langle \phi(x_i), \phi(x)\rangle$, and $\Sigma_o = \left(\frac{1}{2\sigma^2\beta} I_t + K\right)^{-1}$.

To do so, we first observe that the posterior π_t is a multivariate Gaussian.

Lemma 8. *Suppose $\pi = \mathcal{N}(0, \sigma_o^2 I_d)$, $\ell(\cdot, \cdot)$ is the squared-loss, and we have observed $(x_1, y_1), \cdots, (x_t, y_t)$. If take $\Phi \in \mathbb{R}^{t \times d}$ to denote the matrix*

$$\Phi = \begin{bmatrix} — & \phi(x_1) & — \\ — & \phi(x_2) & — \\ & \vdots & \\ — & \phi(x_t) & — \end{bmatrix}.$$

then π_t is the multivariate Gaussian $\mathcal{N}(\widehat{\mu}, \widehat{\Sigma})$ where $\widehat{\Sigma} = \left(2\beta\Phi^T\Phi + \frac{1}{\sigma_o^2} I_d\right)^{-1}$ and $\widehat{\mu} = 2\beta\widehat{\Sigma}\Phi^T y$.

Proof. By expanding the form of $\mathcal{N}(\widehat{\mu}, \widehat{\Sigma})$, we first see

$$\begin{aligned}
\mathcal{N}(w \,|\, \widehat{\mu}, \widehat{\Sigma}) &\propto \exp\left(-\frac{1}{2}(w-\widehat{\mu})^T \widehat{\Sigma}^{-1}(w-\widehat{\mu})\right) \\
&= \exp\left(-\frac{1}{2}w^T\widehat{\Sigma}^{-1}w - \frac{1}{2}\widehat{\mu}^T\widehat{\Sigma}^{-1}\widehat{\mu} + w^T\widehat{\Sigma}^{-1}\widehat{\mu}\right) \\
&\propto \exp\left(-\frac{1}{2}w^T\widehat{\Sigma}^{-1}w + w^T\widehat{\Sigma}^{-1}(2\beta\widehat{\Sigma}\Phi^T y)\right) \\
&= \exp\left(-\frac{1}{2}w^T(2\beta\Phi^T\Phi + \frac{1}{\sigma_o^2}I)w + 2\beta w^T\Phi^T y\right) \\
&= \exp\left(-\beta\left(w^T\Phi^T\Phi w - 2w^T\Phi^T y\right) - \frac{\|w\|^2}{2\sigma_o^2}\right)
\end{aligned}$$

On the other hand, we have

$$\begin{aligned}
\pi_t(w) &\propto \exp\left(-\beta\|y-\Phi w\|^2 - \frac{\|w\|^2}{2\sigma_o^2}\right) \\
&= \exp\left(-\beta\left(y^T y + (\Phi w)^T\Phi w - 2(\Phi w)^T y\right) - \frac{\|w\|^2}{2\sigma_o^2}\right) \\
&\propto \exp\left(-\beta\left(w^T\Phi^T\Phi w - 2w^T\Phi^T y\right) - \frac{\|w\|^2}{2\sigma_o^2}\right)
\end{aligned}$$

Thus $\pi_t = \mathcal{N}(\widehat{\mu}, \widehat{\Sigma})$. □

Since π_t is a multivariate Gaussian, we know that if $w \sim \pi_t$ and $v \in \mathbb{R}^d$ then $\langle w, v\rangle$ is distributed according to $\mathcal{N}(\mu, \sigma^2)$ where

$$\begin{aligned}
\mu &= v^T\widehat{\mu} = 2\beta v^T\widehat{\Sigma}\Phi^T y \\
\sigma^2 &= v^T\widehat{\Sigma}v = v^T\left(2\beta\Phi^T\Phi + \frac{1}{\sigma_o^2}I_d\right)^{-1} v
\end{aligned}$$

Unfortunately, directly computing μ and σ^2 in the forms above requires expanding out the feature mappings, which is undesirable. However, the following theorem, known as the Woodbury Matrix Identity [21], allows us to rewrite these terms in a form only involving inner products of the feature vectors.

Theorem 9 (Woodbury Matrix Identity)**.** *Let T, W, U, V be matrices of the appropriate sizes. Then*

$$(T + UW^{-1}V)^{-1} = T^{-1} - T^{-1}U(W + VT^{-1}U)^{-1}VT^{-1}.$$

Theorem 9 implies that we can rewrite $\widehat{\Sigma}$ as

$$\widehat{\Sigma} = \sigma_o^2\left(I_d - \Phi^T\Sigma_0\Phi\right)$$

where $\Sigma_o = \left(\frac{1}{2\sigma_o^2\beta}I_t + \Phi\Phi^T\right)^{-1}$. With this observation in hand, Lemma 2 follows immediately.

C.3 Proof of Lemma 3

Lemma 3. *Fix any t and suppose the expert provides an answer to atomic question $a_t \in A(q_t)$ at time t. Let $\gamma_t = \pi_{t-1}(\{g \in \mathcal{G} : g(a_t) = g^*(a_t)\})$. Define Δ_t by:*

$$\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}, q_t, a_t\right] \;=\; (1-\Delta_t)\frac{1}{\pi_{t-1}(g^*)},$$

Under Assumption 1, Δ_t can be lower-bounded as follows:

(a) If $\lambda = 1$ (noiseless setting), $\Delta_t \geq (1-\gamma_t)(1-e^{-\beta})$.

(b) For $\lambda \in (0,1)$, if $\beta \le \lambda/2$, then $\Delta_t \ge \beta\lambda(1-\gamma_t)/2$.

Proof. Let $y_1, y_2, \ldots$ denote the possible answers to a_t, and set $p_j = \eta(y_j|a_t, q_t)$ be the probability that the labeler answers y_j. Without loss of generality, suppose $p_1 \ge p_2 \ge \cdots$, so that (under Assumption 1) $g^*(a_t) = y_1$ and $p_1 - p_2 \ge \lambda$.

Further, define $\mathcal{G}_j = \{g \in \mathcal{G} : g(a_t) = y_j\}$. Thus $\gamma_t = \pi_{t-1}(\mathcal{G}_1)$. By averaging over the expert's possible responses, we have

$$\begin{aligned}
&\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}, q_t, a_t\right] \\
&= p_1 \frac{\pi_{t-1}(\mathcal{G}_1) + e^{-\beta}(1-\pi_{t-1}(\mathcal{G}_1))}{\pi_{t-1}(g^*)} + \sum_{j>1} p_j \frac{\pi_{t-1}(\mathcal{G}_j) + e^{-\beta}(1-\pi_{t-1}(\mathcal{G}_j))}{e^{-\beta}\pi_{t-1}(g^*)} \\
&= \frac{1}{\pi_{t-1}(g^*)}\left(p_1(\pi_{t-1}(\mathcal{G}_1) + e^{-\beta}(1-\pi_{t-1}(\mathcal{G}_1)) + \sum_{j>1} p_j(e^{\beta}\pi_{t-1}(\mathcal{G}_j) + (1-\pi_{t-1}(\mathcal{G}_j)))\right) \\
&= \frac{1}{\pi_{t-1}(g^*)}\left(p_1(1-(1-\pi_{t-1}(\mathcal{G}_1))(1-e^{-\beta})) + \sum_{j>1} p_j(1+(e^{\beta}-1)\pi_{t-1}(\mathcal{G}_j))\right) \\
&= \frac{1}{\pi_{t-1}(g^*)} - \frac{1}{\pi_{t-1}(g^*)}\left(p_1(1-\pi_{t-1}(\mathcal{G}_1))(1-e^{-\beta}) - \sum_{j>1} p_j(e^{\beta}-1)\pi_{t-1}(\mathcal{G}_j)\right).
\end{aligned}$$

Setting the parenthesized term to Δ_t, we have

$$\Delta_t \;\ge\; p_1(1-\pi_{t-1}(\mathcal{G}_1))(1-e^{-\beta}) - p_2(e^{\beta}-1)\sum_{j>2}\pi_{t-1}(\mathcal{G}_j) \;=\; \left(p_1(1-e^{-\beta}) - p_2(e^{\beta}-1)\right)(1-\gamma_t).$$

This yields (a) in the lemma statement. For (b), using the inequalities $e^{\beta} \le 1 + \beta + \beta^2$ and $e^{-\beta} \le 1 - \beta + \beta^2$ for $0 \le \beta \le 1$, we get

$$\Delta_t \;\ge\; \left(p_1(\beta-\beta^2) - p_2(\beta+\beta^2)\right)(1-\gamma_t) \;\ge\; \beta\left((p_1-p_2) - \beta(p_1+p_2)\right)(1-\gamma_t) \;\ge\; \beta(\lambda-\beta)(1-\gamma_t).$$

Taking $\beta \le \lambda/2$ completes the proof. □

C.4 Proof of Lemma 4

Lemma 10. *There exists a constant $c > 0$ such that the random variable $U_t = u(q_t; \pi_{t-1})$ satisfies*

$$\mathbb{E}[U_t \,|\, \mathcal{F}_{t-1}] \ge c\,\pi_{t-1}(g^*)(1-\pi_{t-1}(g^*))$$

for any round t where q_t is the query shown to the user at time t and the expectation is taken over the randomness of structural QBC.

Proof. Note that by the structural QBC strategy, the probability that $q \in \mathcal{Q}$ is selected is proportional to $\nu(q)\,u(q;\pi_{t-1})$. This implies

$$\mathbb{E}[U_t \,|\, \mathcal{F}_{t-1}] = \frac{\mathbb{E}_{q\sim\nu}[u(q;\pi_{t-1})^2]}{\mathbb{E}_{q\sim\nu}[u(q;\pi_{t-1})]} \ge \mathbb{E}_{q\sim\nu}[u(q;\pi_{t-1})].$$

Now take $\nu_o = \min_{q\in\mathcal{Q}} \nu(q)$ and $A_o = \max_{q\in\mathcal{Q}} |A(q)|$. Then for any $g \in \mathcal{G}$ s.t. $g \ne g^*$, there exists some $a \in \mathcal{A}$ such that $g(a) \ne g^*(a)$ which implies

$$\mathbb{E}_{q\sim\nu}[d(g,g^*;q)] = \sum_{q\in\mathcal{Q}} \frac{\nu(q)}{|A(q)|} \sum_{a\in A(q)} \mathbf{1}[g(a) \ne g^*(a)] \ge \frac{\nu_o}{A_o}.$$

Then we have

$$\begin{aligned}
\mathbb{E}[U_t \,|\, \mathcal{F}_{t-1}] &\ge \mathbb{E}_{q\sim\nu}[u(q;\pi_{t-1})] \\
&= \mathbb{E}_{g,g'\sim\pi_{t-1}}\left[\mathbb{E}_{q\sim\nu}[d(g,g';q)]\right]
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{g,g'} \pi_{t-1}(g)\pi_{t-1}(g') \left[\mathbb{E}_{q\sim\nu}[d(g,g';q)]\right] \\
&\geq \pi_{t-1}(g^*) \sum_{g\neq g^*} \pi_{t-1}(g)\mathbb{E}_{q\sim\nu}[d(g,g^*;q)] \\
&\geq \frac{\nu_o}{A_o}\pi_{t-1}(g^*)(1-\pi_{t-1}(g^*)).
\end{aligned}
$$

□

Lemma 4. *Suppose that $\mathcal{G}$ is finite and the user's feedback obeys Assumption 2. Then there exists a constant $c > 0$ such that for every round t*

$$\mathbb{E}[1-\gamma_t \,|\, \mathcal{F}_{t-1}] \;\geq\; c\,\pi_{t-1}(g^*)^2(1-\pi_{t-1}(g^*))^2$$

where $\gamma_t = \pi_{t-1}(\{g \in \mathcal{G} : g(a_t) = g^*(a_t)\})$ and a_t is the atom the user provides feedback on.

Proof. Suppose that structural QBC shows the user query q_t. Let $\epsilon > 0$ and $U_t = u(q_t;\pi_{t-1})$ be the random variable denoting the uncertainty of q_t. For an atom $a \in A(q_t)$, we say that a is *known* if

$$\pi_{t-1}(\{g : g(a) \neq g^*(a)\}) < \epsilon\, U_t.$$

And we say that a is *unknown* otherwise. By a union bound, we have

$$\pi_{t-1}(\{g : g(a) \neq g^*(a) \text{ for some known } a \in A(q_t)\}) \leq \epsilon\, U_t |A(q_t)|.$$

By Lemma 1, we have that for any $a \in \mathcal{A}$

$$\pi_{t-1}(\{g : g(a) \neq g^*(a)\}) \geq \frac{1}{2}u(a;\pi_{t-1}).$$

Moreover, there is *some* unknown atom $a^* \in A(q_t)$ with $u(a^*;\pi_{t-1}) \geq U_t$, implying

$$\pi_{t-1}(\{g : g(a^*) \neq g^*(a^*)\}) \geq \frac{1}{2}u(a^*;\pi_{t-1}) \geq \frac{U_t}{2}.$$

So with probability at least $U_t\left(\frac{1}{2} - \epsilon|A(q_t)|\right)$, a random draw from π_{t-1} gets all the known atoms correct and some unknown atom incorrect. Then conditioned on this event occurring, the user has probability at least p_o of correcting some unknown atom. So taking $\epsilon = \frac{1}{4A_o}$ where $A_o = \max|A(q)|$, we have

$$
\begin{aligned}
\mathbb{E}\left[1-\gamma_t \,|\, \mathcal{F}_{t-1}\right] &= \mathbb{E}\left[\pi_{t-1}(\{g : g(a_t) \neq g^*(a_t)\}) \,|\, \mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\pi_{t-1}(\{g : g(a_t) \neq g^*(a_t)\}) | u(q_t;\pi_{t-1}) = U_t\right] \,|\, \mathcal{F}_{t-1}\right] \\
&\geq \mathbb{E}\left[\frac{U_t^2 p_o}{16A_o} \,\middle|\, \mathcal{F}_{t-1}\right] \geq c_o\mathbb{E}[U_t^2 \,|\, \mathcal{F}_{t-1}] \geq c_o\mathbb{E}[U_t \,|\, \mathcal{F}_{t-1}]^2
\end{aligned}
$$

where $c_o = p_o/(16A_o)$ is a positive constant. But by Lemma 10, we know $\mathbb{E}[U_t] \geq c\,\pi_{t-1}(g^*)(1-\pi_{t-1}(g^*))$ for some constant $c > 0$. The lemma follows by substitution. □

C.5 Proof of Lemma 5

Lemma 5. *Let $f : [0,1] \to \mathbb{R}_{\geq 0}$ be a continuous function such that $f(1) = 0$ and $f(x) > 0$ for all $x \in (0,1)$. If*

$$\mathbb{E}\left[\frac{1}{\pi_t(g^*)} \,\middle|\, \mathcal{F}_{t-1}\right] \leq \frac{1}{\pi_{t-1}(g^*)} - f(\pi_{t-1}(g^*))$$

for each $t \in \mathbb{N}$, then $\pi_t(g^*) \to 1$ almost surely.

Proof. Let $X_t = \pi_t(g^*)$. By assumption, $\frac{1}{X_t}$ is a positive supermartingale, which implies $\lim_{t\to\infty}\frac{1}{X_t} = \frac{1}{X}$ exists and is finite with probability one. By the Continuous Mapping Theorem, this implies $\lim_{t\to\infty} X_t = X$ and is non-zero with probability one. On the other hand, we have by the law of total expectation

$$1 \leq \mathbb{E}\left[\frac{1}{X_T}\right] \leq \frac{1}{\pi(g^*)} - \sum_{t=0}^{T-1}\mathbb{E}[f(X_t)].$$

for all $T \in \mathbb{N}$, which implies that $\lim_{t\to\infty} \mathbb{E}[f(X_t)] = 0$. By Fatou's lemma and the Continuous Mapping Theorem, we have

$$0 = \lim_{t\to\infty} \mathbb{E}[f(X_t)] = \mathbb{E}\left[\lim_{t\to\infty} f(X_t)\right] = \mathbb{E}\left[f(X)\right].$$

Thus, $f(X) = 0$ with probability one. Since f has only two potential zeros at 0 and 1, and since $X > 0$ with probability one, we conclude that $X = 1$ with probability one. □

C.6 Proof of Theorem 6

Theorem 6. *Suppose that $\mathcal{G}$ is finite, and Assumptions 1 and 2 hold. If $\pi(g^*) > 0$, then $\pi_t(g^*) \to 1$ almost surely under structural QBC's query strategy.*

Proof. Combining Lemmas 3 and 4, we have

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}\right] &= (1 - \mathbb{E}[\Delta_t|\mathcal{F}_{t-1}])\frac{1}{\pi_{t-1}(g^*)} \\
&\leq \left(1 - \frac{c\beta\lambda\pi_{t-1}(g^*)^2(1-\pi_{t-1}(g^*))^2}{2}\right)\frac{1}{\pi_{t-1}(g^*)} \\
&= \frac{1}{\pi_{t-1}(g^*)} - \frac{c\beta\lambda\pi_{t-1}(g^*)(1-\pi_{t-1}(g^*))^2}{2}
\end{aligned}$$

Now $f(x) = \frac{c\beta\lambda x(1-x)^2}{2}$ meets all of the conditions of Lemma 5, which concludes the proof. □

C.7 Proof of Theorem 7

Our goal is to prove the following theorem.

Theorem 7. *Suppose we are in the general loss setting, $\mathcal{G}$ is finite, and the user's feedback satisfies Assumptions 2, 3, and 4. If $\pi(g^*) > 0$, then $\pi_t(g^*) \to 1$ almost surely.*

To do so, we first give a general loss analogue of Lemma 3.

Lemma 11. *Suppose Assumptions 3 and 4 hold. Take $\beta \leq \min(\lambda/(2C^2), 1/B)$. Suppose that at time t, the user provides feedback on an atomic question $a_t \in A(q_t)$ for which $\mathrm{var}(a_t; \pi_{t-1}) \geq \gamma$. Then*

$$\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}, q_t, a_t\right] \;=\; (1-\Delta_t)\frac{1}{\pi_{t-1}(g^*)},$$

where $\Delta_t \geq \beta\gamma\lambda/2$.

Proof. Plugging in the update rule for the posterior distribution, we have

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}, q_t, a_t\right] &= \sum_y \eta(y|a_t)\frac{\sum_{g\in\mathcal{G}} \pi_{t-1}(g)\exp(-\beta\cdot\ell(g(a_t),y))}{\pi_{t-1}(g^*)\exp(-\beta\cdot\ell(g^*(a_t),y))} \\
&= \frac{1}{\pi_{t-1}(g^*)}\sum_y \eta(y|a_t)\sum_g \pi_{t-1}(g)\exp(-\beta\left[\ell(g(a_t),y) - \ell(g^*(a_t),y)\right])
\end{aligned}$$

(replacing the sum with a suitable integral if necessary.) To turn this expression into the form $(1-\Delta_t)\frac{1}{\pi_{t-1}(g^*)}$, define

$$\Delta_t = \sum_y \eta(y|a_t)\sum_g \pi_{t-1}(g)(1 - \exp(-\beta\left[\ell(g(a_t),y) - \ell(g^*(a_t),y)\right])).$$

Since $\beta \leq 1/B$, and losses lie in $[0, B]$, we have that $\beta[\ell(g(a_t),y) - \ell(g^*(a_t),y)]$ lies in the range $[-1, 1]$. Using the inequality $e^x \leq 1 + x + x^2$ for $-1 \leq x \leq 1$, we then have

$$\Delta_t \geq \sum_g \pi_{t-1}(g)\sum_y \eta(y|a_t)(\beta[\ell(g(a_t),y) - \ell(g^*(a_t),y)] - \beta^2[\ell(g(a_t),y) - \ell(g^*(a_t),y)]^2)$$

$$
\begin{aligned}
&= \sum_g \pi_{t-1}(g) \left(\beta[L(g(a_t), a_t) - L(g^*(a_t), a_t)] - \beta^2 \sum_y \eta(y|a_t)[\ell(g(a_t), y) - \ell(g^*(a_t), y)]^2 \right) \\
&\geq \sum_g \pi_{t-1}(g) \left(\beta\lambda \|g(a_t) - g^*(a_t)\|^2 - \beta^2 \sum_y \eta(y|a_t) C^2 \|g(a_t) - g^*(a_t)\|^2 \right) \\
&= (\beta\lambda - \beta^2 C^2) \sum_g \pi_{t-1}(g) \, \|g(a_t) - g^*(a_t)\|^2 \\
&\geq (\beta\lambda - \beta^2 C^2) \mathrm{var}(a_t; \pi_{t-1}) \;\geq\; \beta(\lambda - \beta C^2)\gamma,
\end{aligned}
$$

where the second and third lines have used Assumptions 3 and 4. □

Next, we lower bound the expected variance of queries provided by structural QBC.

Lemma 12. *Suppose $\mathcal{G}$ is finite. For each round t, the query q_t under the structural QBC strategy satisfies*

$$
\mathbb{E}[\mathrm{var}(q_t; \pi_{t-1}) \,|\, \mathcal{F}_{t-1}] \geq c \, \pi_{t-1}(g^*)(1 - \pi_{t-1}(g^*)).
$$

Proof. For a fixed query q, the probability that q gets chosen in round t can be written as

$$
\Pr(\text{query } q \,|\, \mathcal{F}_{t-1}) = \frac{\nu(q) \, \mathrm{var}(q; \pi_{t-1})}{\sum_{q' \in \mathcal{Q}} \nu(q') \, \mathrm{var}(q'; \pi_{t-1})} = \frac{\nu(q) \, \mathrm{var}(q; \pi_{t-1})}{\mathbb{E}_{q' \sim \nu}[\mathrm{var}(q'; \pi_{t-1})]}.
$$

Taking the expectation of $\mathrm{var}(q; \pi_{t-1})$ over this distribution gives us

$$
\mathbb{E}[\mathrm{var}(q_t; \pi_{t-1}) \,|\, \mathcal{F}_{t-1}] \geq \frac{\mathbb{E}_{q \sim \nu}[\mathrm{var}(q; \pi_{t-1})^2]}{\mathbb{E}_{q \sim \nu}[\mathrm{var}(q; \pi_{t-1})]} \geq \mathbb{E}_{q \sim \nu}[\mathrm{var}(q; \pi_{t-1})].
$$

Now take $\nu_o = \min_{q \in \mathcal{Q}} \nu(q)$ and

$$
d_o = \min_{g \neq g^*} \max_{q \in \mathcal{Q}, a \in A(q)} \frac{\|g(a) - g^*(a)\|^2}{|A(q)|}.
$$

For any particular $a \in \mathcal{A}$,

$$
\begin{aligned}
\mathrm{var}(a; \pi_{t-1}) &= \frac{1}{2} \sum_{g, g' \in \mathcal{G}} \pi_{t-1}(g) \, \pi_{t-1}(g') \, \|g(a) - g'(a)\|^2 \\
&\geq \pi_{t-1}(g^*) \sum_{g \neq g^*} \pi_{t-1}(g) \, \|g(a) - g^*(a)\|^2
\end{aligned}
$$

Which implies

$$
\begin{aligned}
\mathbb{E}_{q \sim \nu} \left[\mathrm{var}(q; \pi_{t-1})\right] &= \sum_{q \in \mathcal{Q}} \frac{\nu(q)}{|A(q)|} \sum_{a \in A(q)} \mathrm{var}(a; \pi_{t-1}) \\
&\geq \sum_{q \in \mathcal{Q}} \frac{\nu(q)}{|A(q)|} \sum_{a \in A(q)} \pi_{t-1}(g^*) \sum_{g \neq g^*} \pi_{t-1}(g) \|g(a) - g^*(a)\|^2 \\
&= \pi_{t-1}(g^*) \sum_{g \neq g^*} \pi_{t-1}(g) \sum_{q \in \mathcal{Q}} \frac{\nu(q)}{|A(q)|} \sum_{a \in A(q)} \|g(a) - g^*(a)\|^2 \\
&\geq \nu_o d_o \pi_{t-1}(g^*)(1 - \pi_{t-1}(g^*))
\end{aligned}
$$

□

We can now give a general loss analogue of Lemma 4.

Lemma 13. *Suppose $\mathcal{G}$ is finite and Assumption 2 holds. Then there exists a constant $c > 0$ such that for any round t*

$$
\mathbb{E}[\mathrm{var}(a_t; \pi_{t-1}) \,|\, \mathcal{F}_{t-1}] \geq c \, \pi_{t-1}(g^*)^3 (1 - \pi_{t-1}(g^*))^2
$$

where a_t is the atom the user provides feedback on and the expectation is taken over both the randomness of user's response and the randomness of structural QBC.

Proof. We first relate the variance of an atom to the probability of a mistake on that atom. To this end, recall

$$D \;=\; \max_{a\in\mathcal{A}} \max_{g,g'\in\mathcal{G}} \|g(a)-g'(a)\|^2.$$

Moreover, since $\mathcal{G}$ and $\mathcal{A}$ are finite, we have the set of realizable values $S = \{g(a) : a \in \mathcal{A}, g \in \mathcal{G}\}$ is also finite. Let $d_o > 0$ denote the minimum squared distance between any two elements of S:

$$d_o \;=\; \min_{\substack{s,s'\in S:\\ s\neq s'}} \|s-s'\|^2.$$

For any atom $a \in \mathcal{A}$, we have

$$\frac{1}{D}\text{var}(a;\pi_{t-1}) \;\leq\; \pi_{t-1}(\{g : g(a) \neq g^*(a)\}) \;\leq\; \frac{\text{var}(a;\pi_{t-1})}{d_o\,\pi_{t-1}(g^*)}.$$

To see the left-hand inequality, we can work out

$$\begin{aligned}
\pi_{t-1}(\{g : g(a) \neq g^*(a)\}) &\geq \frac{1}{2}u(a;\pi_{t-1}) \\
&= \frac{1}{2}\sum_{g,g'} \pi_{t-1}(g)\pi_{t-1}(g')\mathbf{1}[g(a)\neq g'(a)] \\
&\geq \frac{1}{2}\sum_{g,g'} \pi_{t-1}(g)\pi_{t-1}(g')\frac{\|g(a)-g'(a)\|^2}{D} \\
&= \frac{1}{D}\text{var}(a;\pi_{t-1})
\end{aligned}$$

To see the right-hand inequality, notice

$$\begin{aligned}
\text{var}(a;\pi_{t-1}) &= \frac{1}{2}\sum_{g,g'} \pi_{t-1}(g)\pi_{t-1}(g')\|g(a)-g'(a)\|^2 \\
&\geq \pi_{t-1}(g^*)\sum_{g} \pi_{t-1}(g)\|g(a)-g^*(a)\|^2 \\
&\geq \pi_{t-1}(g^*)\sum_{g} \pi_{t-1}(g)\, d_o\, \mathbf{1}[g(a)\neq g^*(a)] \\
&\geq d_o\,\pi_{t-1}(g^*)\pi_{t-1}(\{g : g(a)\neq g^*(a)\})
\end{aligned}$$

Now suppose that structural QBC shows the user query q_t. Let $\epsilon > 0$ and $V_t = \text{var}(q_t;\pi_{t-1})$ be the random variable denoting the variance of q_t. For an atom $a \in A(q_t)$, we say that a has *low variance* if

$$\text{var}(a;\pi_{t-1}) \;<\; \epsilon V_t.$$

And we say that a has *high variance* otherwise. Then by a union bound, we have

$$\pi_{t-1}(\{g : g(a)\neq g^*(a) \text{ for some low variance } a\in A(q_t)\}) \;\leq\; \frac{\epsilon V_t|A(q_t)|}{d_o\pi_{t-1}(g^*)}.$$

On the other hand, there is *some* atom $a \in A(q_t)$ with at least average variance, so that

$$\pi_{t-1}(\{g : g(a)\neq g^*(a)\}) \;\geq\; \frac{\text{var}(a;\pi_{t-1})}{D} \;\geq\; \frac{V_t}{D}.$$

So with probability at least $V_t\left(\frac{1}{D} - \frac{\epsilon|A(q_t)|}{d_o\pi_{t-1}(g^*)}\right)$, a random draw from π_{t-1} gets all the low variance atoms correct and some high variance atom incorrect. Then conditioned on this event occurring, the user has probability at least p_o of correcting some high variance atom. So taking $\epsilon = \frac{d_o\pi_{t-1}(g^*)}{2DA_o}$ where $A_o = \max|A(q)|$, we have

$$\begin{aligned}
\mathbb{E}\left[\text{var}(a_t;\pi_{t-1})\,|\,\mathcal{F}_{t-1}\right] &= \mathbb{E}\left[\mathbb{E}\left[\text{var}(a_t;\pi_{t-1})|\text{var}(q_t;\pi_{t-1}) = V_t\right]\,|\,\mathcal{F}_{t-1}\right] \\
&\geq \mathbb{E}\left[\frac{V_t^2 p_o\epsilon}{2D}\,\middle|\,\mathcal{F}_{t-1}\right] \geq c_o\pi_{t-1}(g^*)\mathbb{E}[V_t^2\,|\,\mathcal{F}_{t-1}] \geq c_o\pi_{t-1}(g^*)\mathbb{E}[V_t\,|\,\mathcal{F}_{t-1}]^2
\end{aligned}$$

where $c_o = \frac{d_o p_o}{4D^2A_o}$ is a positive constant. But by Lemma 10, we know $\mathbb{E}[V_t\,|\,\mathcal{F}_{t-1}] \geq c\,\pi_{t-1}(g^*)(1-\pi_{t-1}(g^*))$ for some constant $c > 0$. The lemma follows by substitution. □

The proof of Theorem 7 now follows exactly in the same fashion as the proof of Theorem 6.

D Convergence rates

In this section, we bound the rate at which structural QBC's posterior concentrates on the target structure g^*. Before we do so, we first introduce some concepts related to the informativeness of feedback.

D.1 Quantifying the informativeness of feedback

For an atomic question a and corresponding answer y, let $\mathcal{G}_{a,y} = \{g \in \mathcal{G} : g(a) = y\}$. Define the *shrinkage* of posterior π' due to atomic question a to be

$$S(\pi, a) = 1 - \max_y \pi(\mathcal{G}_{a,y})$$

and define the shrinkage of π' due to a query q to be the average shrinkage due to q's atoms

$$S(\pi, q) = \frac{1}{|A(q)|} \sum_{a \in A(q)} S(\pi, a).$$

This is very similar to the notion of *information gain* from the original query-by-committee analysis [16], as we explain in further detail at the end of this section. The reason for the modification is that it facilitates the generalizations introduced here: the multiclass setting and noisy feedback.

In Section 4, we saw that by making relatively weak assumptions on the specific atoms a user will provide feedback on when presented with a query, we can guarantee the consistency of structural QBC. To get meaningful convergence rates, however, we will need to make a stronger assumption. Specifically, we will require that when presented with a query q, the user provides feedback on an atom $a \in A(q)$ whose shrinkage is close to the average shrinkage of the atoms in $A(q)$.

Assumption 5. *When shown q_t and $g_t(q_t)$, the user provides feedback that satisfies*

$$\mathbb{E}[S(\pi_{t-1}, a_t)] \;\geq\; S(\pi_{t-1}, q_t)$$

where the expectation is taken over the randomness of the user's choice of a_t.

Later in this section, we characterize the shrinkage that might be expected in a few cases of interest. For the time being, we will think of it as one of the key quantities controlling the efficacy of active learning, and give rates of convergence in terms of it.

D.2 Rate of convergence

In this section, we again consider the setting in which $\mathcal{G}$ is finite. To keep things simple, we think of convergence as occurring when $\pi_t(g^*)$ exceeds some particular threshold τ (such as $1/2$).

Theorem 14. *Suppose that $\mathcal{G}$ is finite, and that the expert's responses satisfy Assumptions 1 and 5. Suppose moreover that there are constants $0 < \tau, s_o < 1$ such that at any time t, if $\pi_t(g^*) \leq \tau$, the shrinkage of the next query is bounded below in expectation as*

$$\mathbb{E}[S(\pi_{t-1}, q_t) | \mathcal{F}_{t-1}] \geq s_o.$$

Pick any $0 < \delta < 1$. With probability at least $1 - \delta$, the number of rounds T of querying before $\pi_T(g^*) > \tau$ can be upper-bounded as follows.

$$T \;\leq\; \begin{cases} \frac{2}{s_o(1-e^{-\beta})} \max\left(\ln \frac{1}{\pi(g^*)}, \frac{4}{s_o(1-e^{-\beta})} \ln \frac{1}{\delta}\right) & \text{if } \lambda = 1 \text{ (noiseless case)} \\ \frac{4}{\beta \lambda s_o} \max\left(\ln \frac{1}{\pi(g^*)}, \frac{8e^{\beta}}{\beta \lambda s_o} \ln \frac{1}{\delta}\right) & \text{for any } \lambda, \text{ if } \beta \leq \lambda/2 \end{cases}$$

Proof. We will spell out the argument for the noisy case; the other case is similar but slightly simpler. Define

$$R_t = 1 - \frac{\pi_{t-1}(g^*)}{\pi_t(g^*)}.$$

Using the random variable Δ_t from Lemma 3, we have

$$\mathbb{E}[R_t | \mathcal{F}_{t-1}] = 1 - \pi_{t-1}(g^*)\, \mathbb{E}\left[\frac{1}{\pi_t(g^*)} \middle| \mathcal{F}_{t-1}\right]$$

$$
\begin{aligned}
&= 1 - \pi_{t-1}(g^*)\,\mathbb{E}\left[\mathbb{E}\left[\frac{1}{\pi_t(g^*)}\middle|\mathcal{F}_{t-1}, q_t, a_t\right]\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}[\Delta_t|\mathcal{F}_{t-1}] \\
&\geq \frac{1}{2}\beta\lambda\,\mathbb{E}[1-\gamma_t|\mathcal{F}_{t-1}] \\
&\geq \frac{1}{2}\beta\lambda\,\mathbb{E}[S(\pi_{t-1}, q_t)|\mathcal{F}_{t-1}] \;\geq\; \frac{1}{2}\beta\lambda s_o.
\end{aligned}
$$

as long as $\pi_{t-1}(g^*) \leq \tau$ throughout.

Pick any time T before $\pi_T(g^*)$ exceeds τ. Then $\mathbb{E}[R_1 + \cdots + R_T] \geq \beta\lambda s_o T/2$. To show that this sum is concentrated around its expected value, we can use a martingale large deviation bound. First we check that each R_t is bounded. Since

$$
e^{-\beta}\pi_{t-1}(g^*) \leq \pi_t(g^*) \leq e^{\beta}\pi_{t-1}(g^*),
$$

it follows that R_t lies in an interval of size at most e^{β}. By the Azuma-Hoeffding inequality [22, 5], if T attains the value in the theorem statement, then with probability at least $1-\delta$,

$$
R_1 + \cdots + R_T \;>\; \frac{1}{2}\mathbb{E}[R_1 + \cdots + R_T] \;\geq\; \frac{1}{4}\beta\lambda s_o T \;\geq\; \ln\frac{1}{\pi(g^*)}.
$$

But this is not possible, since $R_1 + \cdots + R_T$ can be at most $\ln(1/\pi(g^*))$ by the chain of inequalities

$$
1 \;\leq\; \frac{1}{\pi_T(g^*)} \;=\; (1-R_1)(1-R_2)\cdots(1-R_T)\frac{1}{\pi(g^*)} \;\leq\; \exp(-(R_1 + \cdots + R_T))\frac{1}{\pi(g^*)}.
$$

□

It is likely that the quadratic dependence on s_o, λ, and β can be reduced by a more careful large deviation bound.

D.3 Shrinkage and uncertainty

The active learning literature has introduced a variety of different complexity measures that attempt to capture the number of queries needed for learning. These include the *disagreement coefficient* [20] and the aforementioned *information gain* [16]. Although it is possible to give general bounds in terms of these quantities, it has proved quite difficult to compute these complexity measures for all but a few simple cases.

Our notion of shrinkage is a reformulation of the information gain that avoids assuming that the target structure is drawn from the prior distribution, and that accommodates scenarios beyond binary classification. By way of illustration, we will give an example of a simple situation in which the shrinkage can be characterized.

The following lemma demonstrates that under the structural QBC strategy, we can relate a query's shrinkage to its uncertainty.

Lemma 15. *Suppose the current posterior is π_{t-1}. Under Assumption 5, the shrinkage S_t of the user's next response has expected value*

$$
\mathbb{E}[S_t] \;\geq\; \frac{\mathbb{E}_{q\sim\nu}[u(q;\pi_{t-1})^2]}{\mathbb{E}_{q\sim\nu}[u(q;\pi_{t-1})]}.
$$

Proof. Fix any query $q \in \mathcal{Q}$ and atom $a \in A(q)$. From Lemma 1, we have

$$
S_t = \min_y \pi(\{g : g(a) \neq y\}) \geq u(a;\pi_t)/2.
$$

Now, if the next query is q_t, then under Assumption 5,

$$
S_t \;\geq\; \mathbb{E}_{a\sim\text{unif}(A(q_t))}\frac{1}{2}u(a;\pi_t) \;=\; \frac{1}{2}u(q_t;\pi_t).
$$

The lemma follows by taking expectation over $q_t \sim \nu$. □

We now turn to an example where we can bound the shrinkage.

Example: Partitioning a hypercube by axis-parallel cuts

Let $\mathcal{X}$ be the hypercube $[0,1]^p$, and consider axis-aligned bipartitions of $\mathcal{X}$. Any such partition is specified by a coordinate i and a value v, and yields clusters

$$\{x : x_i \leq v\} \text{ and } \{x : x_i > v\}.$$

Let π be the uniform distribution over such partitions $\mathcal{G} = [p] \times [0,1]$, so that $\pi(i,v) = 1/p$; and suppose that the data distribution is uniform over $\mathcal{X}$.

We will take atomic queries to be of the form $\{x, y\}$ for $x, y \in [0,1]^p$, where the answer is 1 if they lie in the same cluster, and 0 otherwise. The following lemma shows that the shrinkage of structural QBC queries is always constant in this setting.

Lemma 16. *Under Assumption 5, the shrinkage S_t of a user's feedback satisfies $\mathbb{E}[S_t] \geq 1/3$.*

Proof. Note that if we query $\{x, y\}$ and they are in separate clusters, the version space shrinks to the regions between x_i and y_i on each coordinate i; while if they are in different clusters, we get the complement. Either way, the resulting version space can be mapped back to the original $(\mathcal{G}, \pi)$ by transforming the data, and hence this is the only case we need consider in computing uncertainty and shrinkage values.

For any $x, y \in [0,1]^p$, the probability that they are separated by a random draw from π is exactly

$$\sum_{i=1}^{p} \Pr(\text{cut coordinate is } i)\, |x_i - y_i| \;=\; \frac{\|x-y\|_1}{p}.$$

Thus the uncertainty on a query $\{x, y\}$ is

$$u(\{x,y\}) \;=\; \Pr_{g,g\sim\pi}(\text{exactly one of } g, g' \text{ separates } x, y) \;=\; 2 \cdot \frac{\|x-y\|_1}{p}\left(1 - \frac{\|x-y\|_1}{p}\right).$$

We will compute the expectation of this over $X = (X_1, \ldots, X_p)$ and $Y = (Y_1, \ldots, Y_p)$ drawn uniformly at random from $[0,1]^p$.

First, a simple one-dimensional calculation shows that

$$\mathbb{E}[\|X - Y\|_1] \;=\; \sum_{i=1}^{p} \mathbb{E}|X_i - Y_i| \;=\; \frac{p}{3}.$$

Likewise,

$$\mathbb{E}[\|X - Y\|_1^2] \;=\; \sum_{i=1}^{p} \mathbb{E}|X_i - Y_i|^2 + \sum_{i \neq j} (\mathbb{E}|X_i - Y_i|)(\mathbb{E}|X_j - Y_j|) \;=\; \frac{p}{6} + \frac{p(p-1)}{9}.$$

Inserting these into the expression for uncertainty, we get

$$\mathbb{E}[u(\{X,Y\})] \;=\; 2\left(\frac{\mathbb{E}\|X-Y\|_1}{p} - \frac{\mathbb{E}\|X-Y\|_1^2}{p^2}\right) \;=\; \frac{4}{9} - \frac{1}{9p} \;\geq\; \frac{1}{3}.$$

We finish by invoking Lemma 15 and observing that $\mathbb{E}[S_t] \geq \mathbb{E}[u(\{X,Y\})^2]/\mathbb{E}[u(\{X,Y\})] \geq \mathbb{E}[u(\{X,Y\})]$. □

D.4 Relation to information gain

The original analysis of query-by-committee was specifically for active learning of binary classifiers and was based on the notion of *information gain* [16]. Suppose the current posterior distribution over classifiers is π, and that under this posterior, a specific query x has probability p of having a positive label and probability $1-p$ of having a negative label. Then the information gain is defined to be the entropy of a coin with bias p,

$$I(\pi, x) = H(p).$$

In this same situation, the shrinkage is

$$S(\pi, x) = 1 - \max(p, 1-p).$$

These two quantities are related by a monotonic transformation. The analysis of QBC's query complexity assumes that the *expected* information gain, taken over the random choice of next query, is always bounded below by a constant. In the analysis presented in this section, we assumed the same of the expected shrinkage. In the case of binary classification, these two conditions coincide.

For instance, [16] showed that if the classifiers are homogeneous (through-the-origin) linear separators, and the data distribution is uniform over the unit sphere, then the expected information gain due to a label query is bounded below by a constant. This means that the same holds for the expected shrinkage.